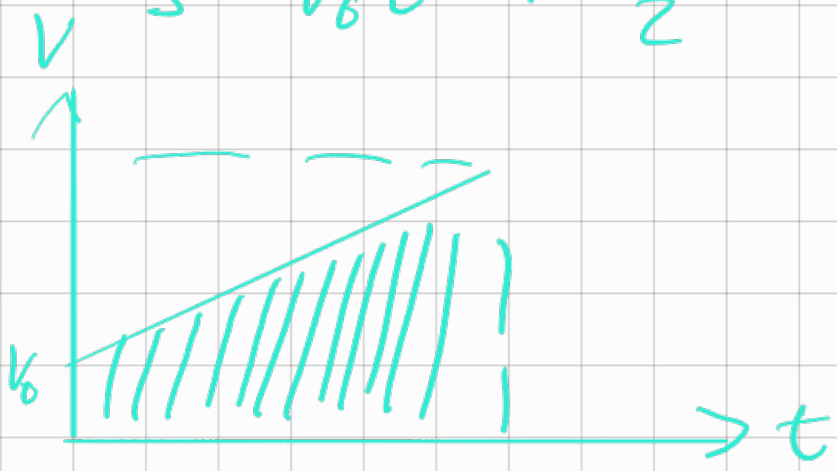
$$x = x_0 + \underbrace{v_0 t + \frac{at^2}{2}}_{S_x}$$

$$v = v_0 + at$$

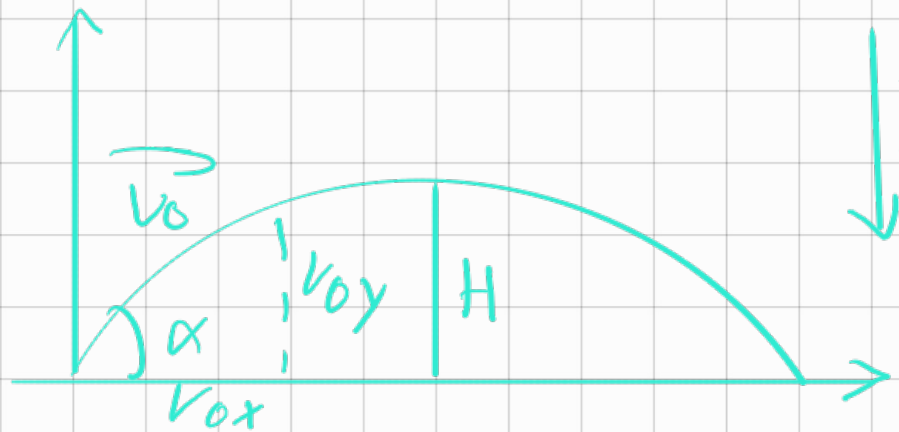
$$S = v_0 t + \frac{at^2}{2}$$



$$S = \frac{(v_0 + v)(v - v_0)}{2a} = \frac{v^2 - v_0^2}{2a}$$

(5) – 1

(7) – 2Ч



$x =$

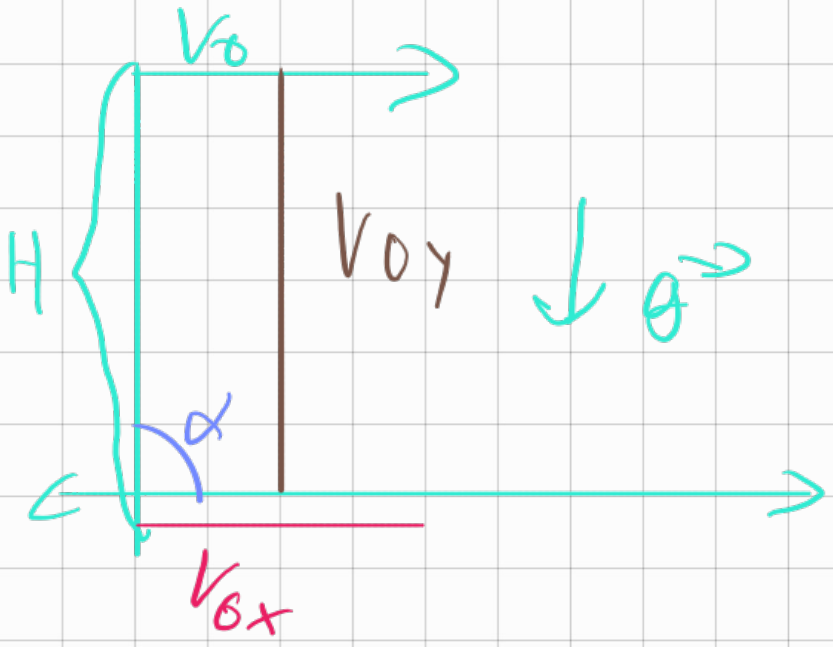
$v_x =$

$y =$

$v_y =$

$V_{0x} = V_0 \cos\alpha$; $V_{0y} = V_0 \sin\alpha$

$y = V_0 \sin\alpha\, t$; $x = V_0 \cos$

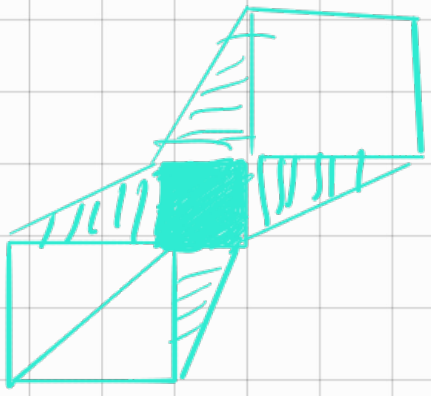


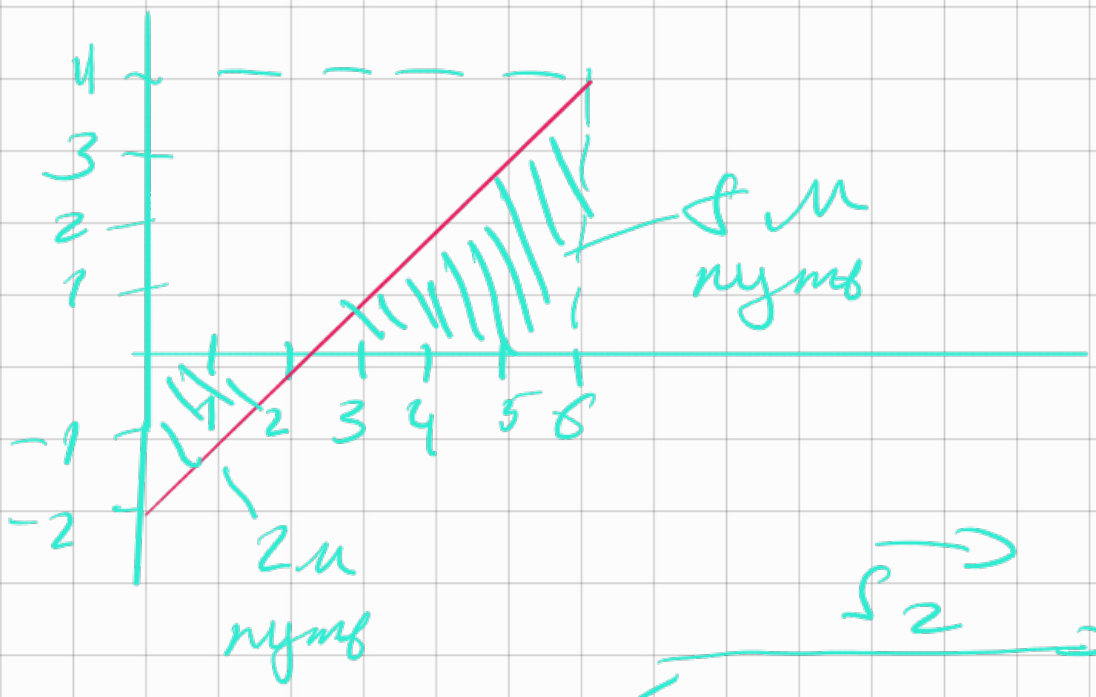
$x = V_0 t$

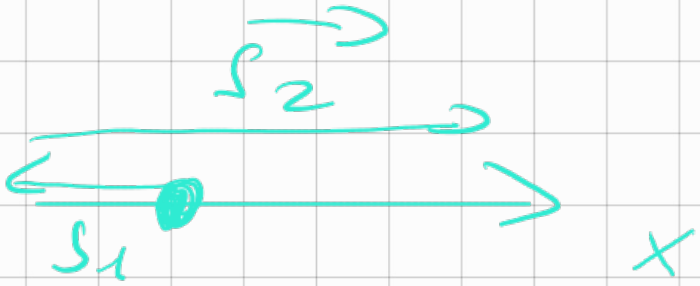
$V_x = V_0$

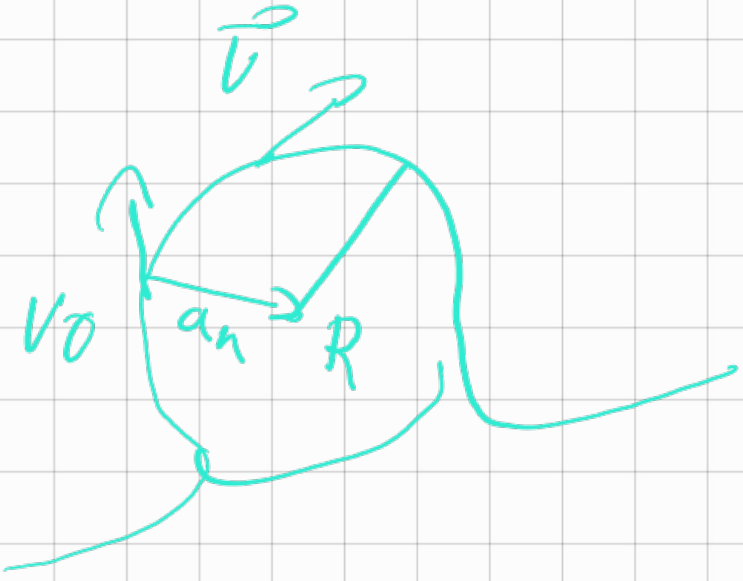
$y = h - \frac{gt^2}{2}$

$V_y = -gt^2$









по модулю

$\vec{v}$

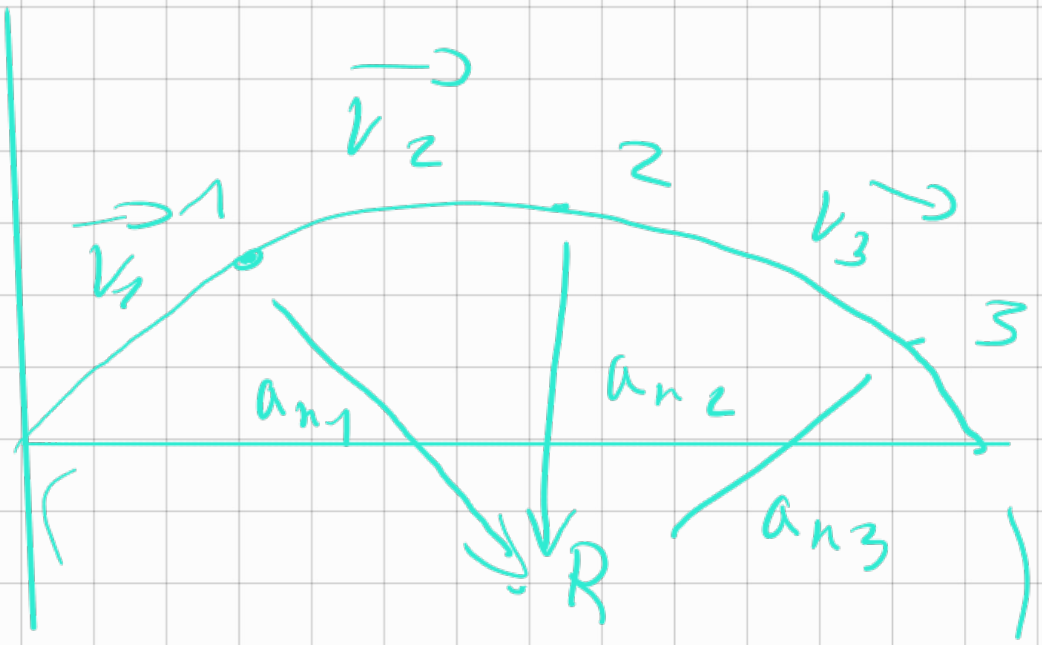
по напр

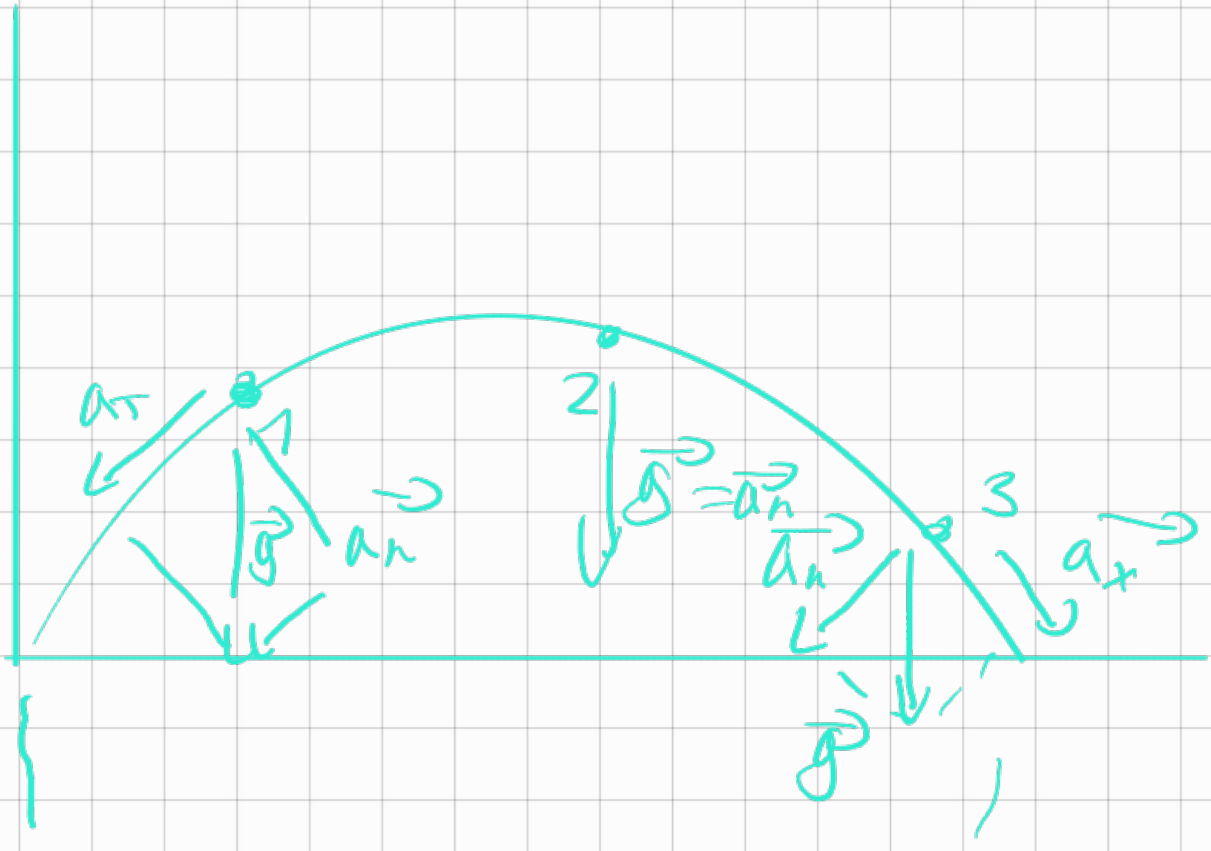
$a_\tau = \frac{dv}{dt}$

$a_n = \frac{v^2}{R}$

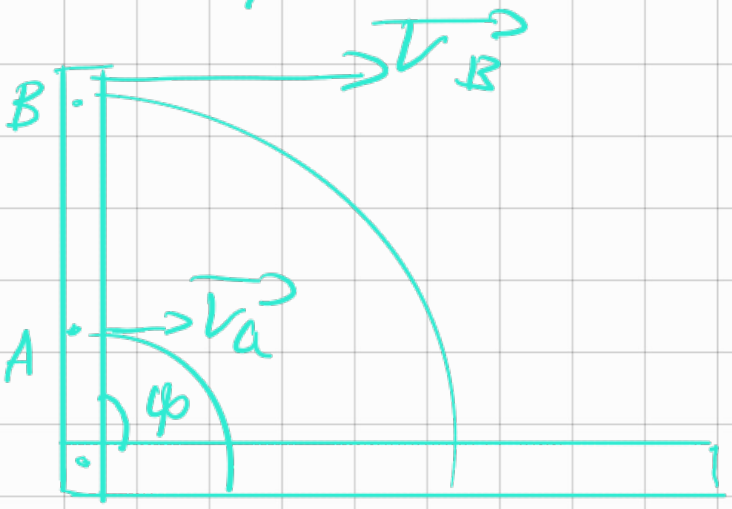
R – радиус кривизны

$$\vec{a} = \vec{a}_\tau + \vec{a}_n$$





ВРАЩАТЕЛЬНОЕ ДВИЖЕНИЕ



v_A, v_B – линейная скорость

поступательное	вращательное
S	φ – угол поворота тела $[\varphi] = рад$
$\vec{v} = \frac{\vec{dS}}{dt}$	$\omega_\varphi = \frac{\Delta\varphi}{\Delta\tau}$ $[\omega] = \frac{рад}{с}$

$$\vec{\varepsilon} = \frac{\Delta \vec{\omega}}{\Delta t}$$

$$(\beta)_n = \frac{\Delta \vec{\omega}}{\Delta t}$$

$$\vec{\varepsilon} = \frac{\Delta \vec{\omega}}{\Delta t}$$

$$\vec{v} = \vec{\omega} \vec{R}$$

$$a_\tau = \varepsilon R$$

$$a_n = \omega^2 R$$

φ — $\Delta\varphi$

$$\varphi = t^3 - 2t^2 + 3; \quad R = 20\ см$$

сро

$$\omega = \frac{d\varphi}{dt} = 3t^2 - 4t$$

$$\omega(t = 1c) = 3 - 4 = -1 \frac{рад}{с}$$

$$\varepsilon = \frac{d\omega}{dt} = 6t - 4$$

$$\varepsilon\ (t = 1c) = 2 \frac{рад}{с}$$

$$v(t = 1c) = \omega \cdot R = -1 \cdot 0{,}2 = -0{,}2 \frac{м}{с}$$

$$a_\tau = \varepsilon R = 2 \cdot 0{,}2 = 0{,}4 \frac{м}{с^2}$$

$$a_n = 1 \cdot 0{,}2 = 0{,}2 \frac{м}{с^2}$$

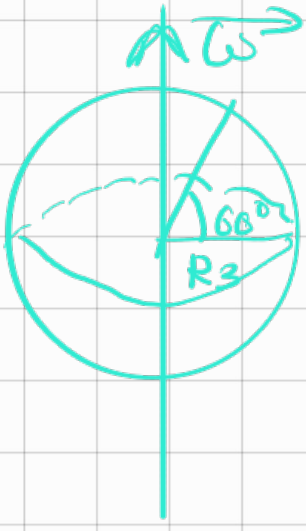
Период имеет смысл при равномерном движении.

T – период $[T] = c$

ν – частота $[\nu] = \frac{1}{c} = [Гц]$

$\nu = \frac{1}{T}$

$\omega = \frac{\Delta\varphi}{\Delta t} = \frac{2\pi}{T} = 2\pi\nu$



$\varphi = 2\pi$

$T = 24\ ч$

$R = 6400\ км$

$V_{СПБ} = \omega \cdot R_{СПБ}$

$R_{СПБ} = \frac{R}{2}$

$\omega = \frac{\Delta\varphi}{\Delta t} = \frac{2\pi}{T}$

$V = \frac{2\pi}{T} \cdot \frac{R}{2} = \frac{\pi R}{T}$

$V_{СПБ} = 232\ \frac{м}{с}$